Course Title: Electric Drive

Code & No : ELE 2461

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At the Beginning

- Textbook is the core
- Lectures for discussions
- Sequential reading
- Solve the problems
- Avoid Stereotyping
- Ask me why and how?

Mistakes

- Reading only the slides
- Compressive reviewing before the exam
- Missing the lectures

Course Material

- G.K. Dubey , Fundamentals of Electric Drives , Narose Publishing House, second edition,2002

- References
- Lecture notes
- Internet

Chapter one

- **Concepts of the Electric Drive System and its classification**

Concepts of Electric Drive System

Electric Drive System(EDS) is the study of the electric system involving controlling electric motors in both steady state and dynamic operation. It is achieved through taking into account the characteristics of the mechanical loads and the behavior of the power electronic converters. **Main functions of EDS are:**

1. Provide transition condition (starting, braking, reverse,..etc)
2. Speed control with specific static error
3. Adaptive control which means that the drive system changes its structure or algorithm to achieve the maximum efficiency .
4. protection: shut down the system in case of emergency or short circuits.

Advantages of EDS:

1. Wide range of torque, speed and power. 2. High efficiency.
3. Flexible control characteristics to meet the load requirements
4. Adaptive to any operating conditions such as explosive, radioactive environment, vertical mountains and so on. 5. Do not pollute the environment
6. Can work in all four quadrants or speed- torque plane.
7. No need for refuel or warm-up the motor. 8. low maintenance

Choice of electric drive

Factors that EDS depends on are:

1.steady –state operation requirements such as nature of speed torque characteristics, speed regulation, speed range, efficiency, duty cycle, quadrants of operation, speed fluctuations ,ratings,…

2. Transient operation requirements such as values of acceleration and deceleration, starting, braking and reversing performance.

3. Requirements related to the source such as type of source and its capacity, magnitude of voltage, voltage fluctuations, power factor, harmonics and its effect on the loads, ability to accept regenerative power.

4. Capital and running cost, maintenance need, life.

5. Space and weight restrictions

6. Environment and location

7. reliability

Classification of drive systems

-According to the purpose:

1- main drive: provides load machine with main motion

2- auxiliary drive : provides load machine with aux. motion

- According to the motion type:

1-rotational 2- translational

-According to the direction of motion:

1-reversible (two direction) 2- irreversible (one direction)

- According to the principle of work:

1- Continuous 2- discrete

Classification of drive systems

- **According to the type of connection with loads:**

1-direct connection 2- connection via reduction means such as gear or belt

- **According to number of motions of load machines**:

1-individual drive 2- group drive 3. interconnected (load is driven by two motors or more)

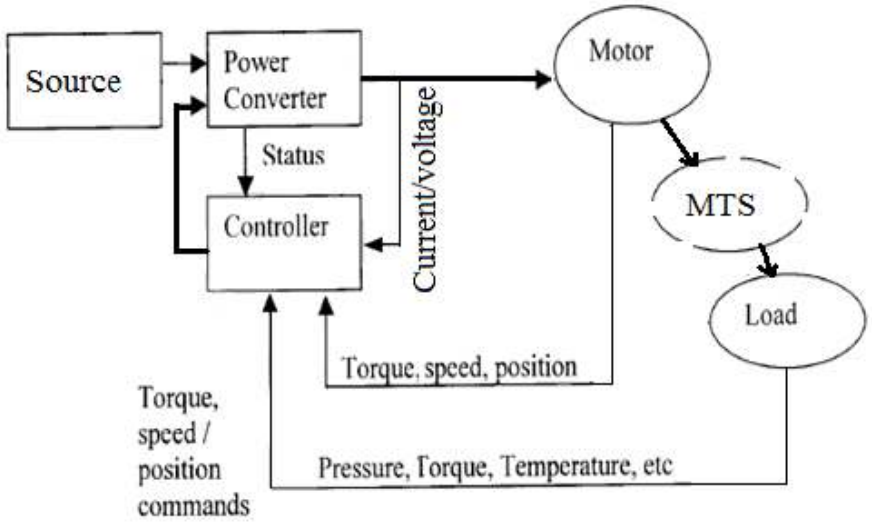
- **According to the feeding current:**

1-ac drive 2-dc drive

- **According to the ability of speed regulation**:

1- regulated 2-unregulated

Components of the Electric Drive System
Block diagram of the electric drive system.



A modern electric drive system has four components:

1. Electric Motor
2. Power Converter: Rectifiers, Choppers, Inverters, Cycloconverters
3. Controllers – matching the motor and power converter to meet the load requirements
4. Load

Some loads are coupled with the motor shaft through transmission mechanical system (TMS).

Motors

Types of electric motors recently used in electric drive systems are:

1. DC motors: Shunt, Series, Compound, Separately excited, Switched reluctance motors
2. AC motors: Induction, Wound rotor , Synchronous, Permanent magnet synchronous and Reluctance motors.
3. Special motors: Switched reluctance motors.

Motor is selected according to the power level, environmental factors and performance required by the load. Ex: if load requires high starting torque so dc series motor is better than induction

Comparison between Dc and AC drives

	DC Drives	AC Drives (particularly Induction Motor)
Motor	• requires maintenance • heavy, expensive • limited speed (due to mechanical construction)	• less maintenance • light, cheaper • high speeds achievable (squirrel-cage IM) • robust
Control Unit	Simple & cheap control even for high performance drives • decoupled torque and flux control • Possible implementation using single analog circuit	Depends on required drive performance • complexity & costs increase with performance • DSPs or fast processors required in high performance drives
Performance	Fast torque and flux control	Scalar control - satisfactory in some applications Vector control - similar to DC drives

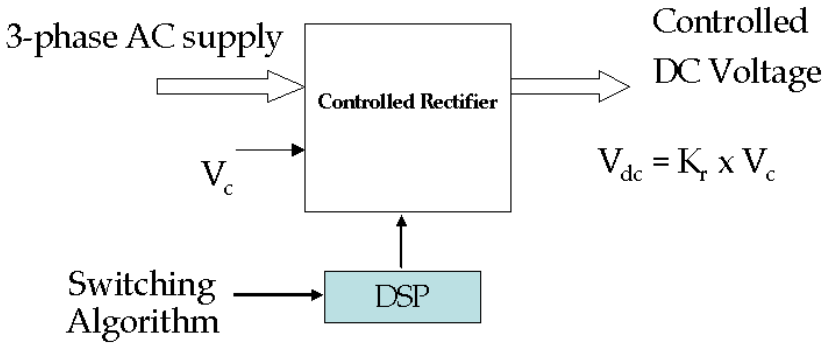
Power Converters

interfaces the motor with the power source and provides the motor with adjustable voltage, current and/or frequency .

The power converters driving the motors are:

1. Controlled Rectifiers (ac to dc)

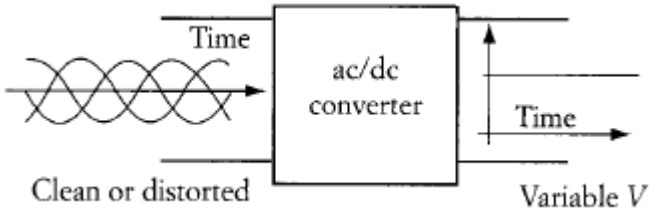
Controlled rectifiers are fed from single and three-phase AC main supply and provide a dc output with adjustable (fixed or variable) magnitude for control of the dc machines or some times input dc supply to the inverters in the case of ac machines..



Where:

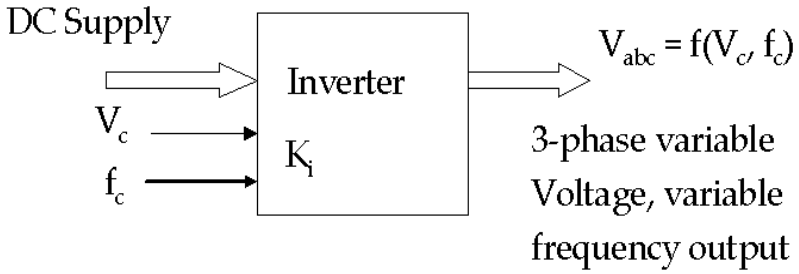
V_c-Voltage controlled

K_r-Gain (proportionality constant)

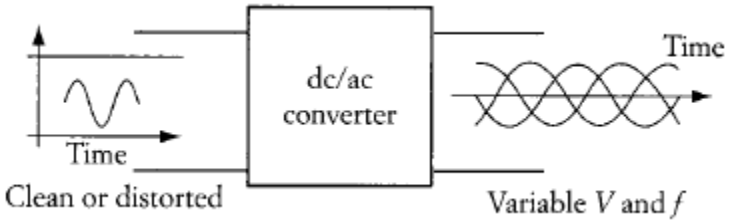


Power Converters

2. Inverters (dc to ac) – Voltage and current source converters are fed from a DC link. The DC link is generated with either a controlled or uncontrolled rectifier. They provide variable alternating voltages and currents at desired frequency and phase for the control of ac machines. This type of converter is suitable for ac motors, such as induction or synchronous motors

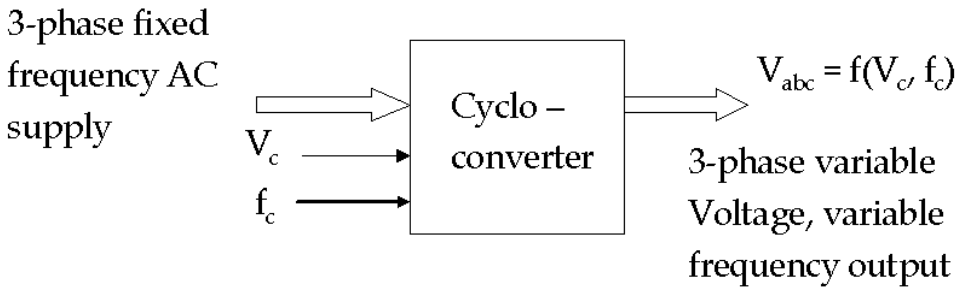


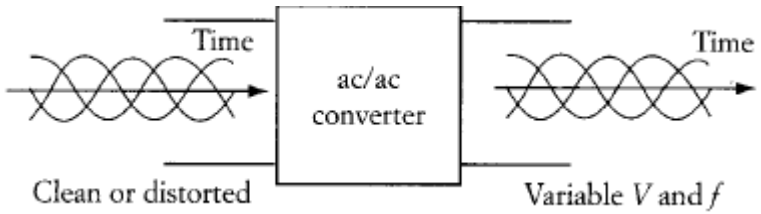
V_c: controlled magnitude command
f_c: frequency command



Power Converters

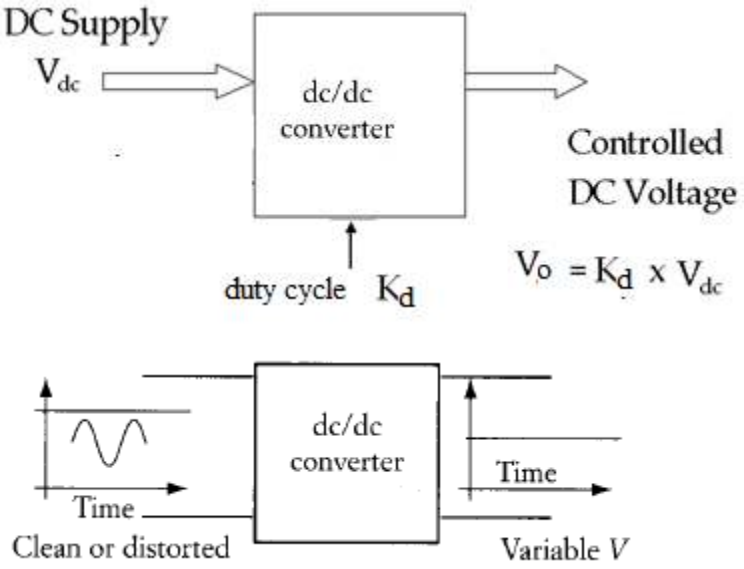
3. Cycloconverter (ac to ac)
They provide a direct conversion of fixed-frequency alternating voltage/current to an ac output with variable magnitude, frequency or both for the control of ac machines. The output frequency is usually limited from 33 to 50% of the input supply frequency, to avoid distortion of the wave form, and therefore they are used only in low-speed but high-power ac motor drives.





Power Converters

4. dc to dc. This type is also known as "chopper". The constant- input dc waveform is converted to a dc waveform with variable magnitude. The typical application of this converter is in dc motor drives.

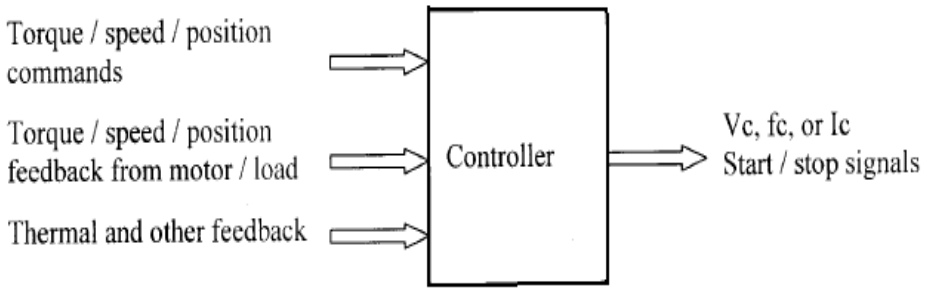


Controllers

supervises the operation of the whole system to ensure stability and enhance the overall performance.

- The controllers implement the control strategy governing the load and motor characteristics
- To match the load and motor, the input to the power converter is controlled (manipulated) by the controller . The inputs to the controller consists of:

1. Torque, flux, speed, and/or position commands
2. Their rate of variations to facilitate soft start, to preserve the mechanical integrity (السلامة) of the load
3. The actual values of torque, flux, speed, and/or position for feedback control.
4. Limiting values of currents, torque, acceleration, etc.
5. Temperature feedback, instantaneous currents and/or voltages in the motor and/or converter.
6. The constants in the speed and position controllers such as proportional, integral and differential gains.



Vc: controlled voltage
fc: controlled frequency
Ic: controlled current

Controllers

• The controller output determines the control signal for voltage magnitude, Vc in the case of inverters, and the control signal for determining the frequency, fc.
• These functions can be merged, and only the final gating signals might be directly issued (ترسل)to the bases/gates of the power converter.
• It may also perform the protection and other monitoring functions and deal with emergencies such as sudden field loss or power failure.
•The controllers are realized with analog and integrated circuits.
• The present trend is to use microprocessors, single chip micro controllers, digital signal processors [DSPs].
• Very-large-scale integration [VLSI], and special custom chips also known as application specific ICs [ASICs] to embody(تجسيد) a set of functions in the controller.
•The real time computational capability of these controllers allows complex control algorithms to be implemented.

Controllers

The drive system performs the following functions:

1- ON-OFF Control : Switching on-off or performing automatic Starting, stopping, and reversing electric motors using different principles.

2- Parametric Control : such as

- Motor Speed Control, the control system can be open-loop or closed- loop.
- Position control (Servo-drive system), It is a closed-loop system with position feedback signal.

3- Adaptive control system, is a programmed closed-loop system with feedback signals, and includes an adaptive unit, and one of its characteristics is that it adapts to any changes that may occur to the system that cannot be predicted in advance. Such a turning machine must adapt to changing the type and hardness of the metal.

4- Optimal control system, the aim of using this system is to minimize (save) energy consumption and improve the performance and efficiency of the drive system in the event (حالة)that the driving system operates at light loads, a load less than the nominal load.

Controllers

Drive System Specifications: The drive system must have the following specifications:

1. Carry out the required function with complete accuracy, high productivity, and without faults (malfunctions).
2. Ease of installation, use, maintenance and high reliability.
3. Provides means of protection for machines and people and provides warning signals.
4. Economical in terms (من حيث) of low price and high efficiency

Power supply

Provides the energy to the drive system

- DC supply – used in some systems such as underground traction with 500-750 V
- AC supply (single phase and three phase)- where single phase is used for very low power drives except in the case of traction drives (even at very high power levels) due to the economy. Rest of other drives are powered from 3-phase power drive
- Battery - used for some applications such as fork lift, milk vans and electric vehicles (Evs)
- Renewable energy such as solar and wind energy – used in some drives such as space and water pumping applications.

Load

The main (fundamental) component of electric drive system is a mechanical tool (industrial machinery). The selection of other components of drive system depends on the mechanical tool parameters, specifications and characteristics. Industrial machinery types: The industrial machineries used in electric drive classified into three groups that have certain common technical specifications.

A) Transporting Machineries, such as:

- Lift (Elevator), used to vertically lift the passengers and load.
- Crane, which is used in buildings and ports to transport loads and goods.
- Escalator used for transporting people between two levels at an angle.
- Conveyor used for transporting materials using belts such as production line.
- The transporting machineries deal with transportation of masses, that have weights depending on gravity acceleration. So, the industrial machineries load torque is active torque and transporting machineries are called "Constant load torque machineries".

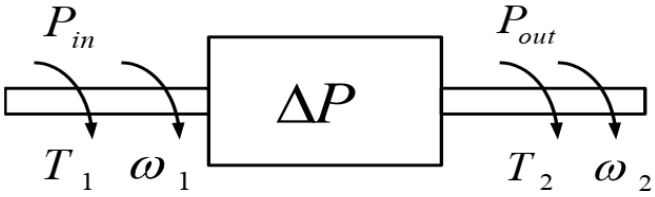
Load

B) Production machineries, which are raw materials (wood, plastic and metal) manufacturing machines, including drilling machines, cutting and milling, turning, sawing, and others. The manufacturing process is a result of direct contact between work-piece and tool, and a contact friction force will produce reactive torque. The speed of tool is inversely proportional with torque variation. The production machineries are called " Constant load power machineries".

C) Ventilation, fan, pump and compressor machineries, which have reactive torque as a result of friction force. The load torque is directly proportional with speed squared. These machineries are called " Fan load characteristic machineries".

Transmission unit

It converts mechanical power into mechanical power, as shown in Figure



And the transmission unit efficiency is equal to: $\eta = \frac{\mathbf{P_{out}}}{\mathbf{P_{in}}} < 1$

The transmission is either geared down (step down) or geared up (step up). In the drive system, only step down transmissions are used to increase torque and decrease speed. That is, we get more torque on the output of the transmission, so it is possible to choose an electric motor at the same power with high speed and low torque, that is, with less dimensions and at a low price. **Three Types of transmissions devices**:

1.gearbox that uses different types of gear, including:

- The cylindrical type, as shown in Figure (1-3).

Transmission unit

- Transmission gear ratio is equal to

$$i = \frac{n_1}{n_2} = \frac{Z_2}{Z_1}$$

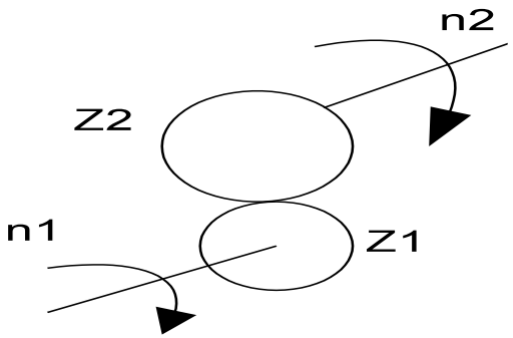


Figure (1-3) a cylindrical type transmission

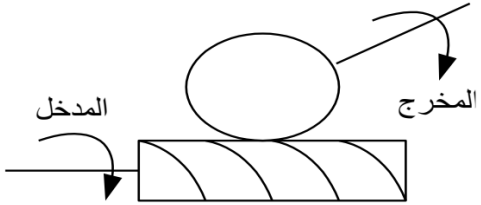
where

i : Gear Ratio, which must be an integer number.

Z : Number of teeth, n1and n2: input and output speed .

- The Worm Gear as shown in Figure (1-4).

This type is characterized by its less dimensions and high efficiency, so it is commonly used in elevators.



Figure(1-4) Worm Gear

Transmission unit

2. Belt drive or chain drive. It is used to transmit the movement at separate distances between the spindles and the transmission device, either by belt drive or chain drive, as in Figure (1-5).

In order for the belt or chain not to be interrupted, the value of linear velocity must be the same for all parts of the belt or chain, that is, that V1 =V2 , and this means that $\omega_1 R_1 = \omega_2 R_2$. From this relationship, we notice that increasing the pulley radius leads to a decrease in the angular velocity.

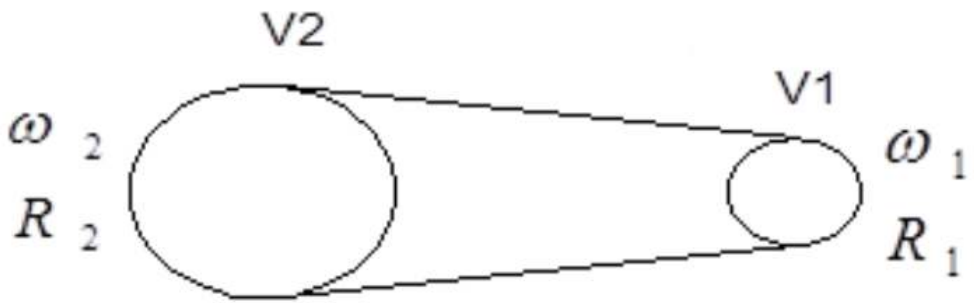


Figure (1-5) Pulley type transmission

Transmission unit

3- Continuously Variable Transmission (CVT): CVT systems offer seamless (سلس بدون طبقات)and continuous variation of the gear ratio between the electric motor and the load. They utilize mechanisms such as belts, pulleys, or hydrostatic transmissions to provide infinite variability in the transmission ratio. CVTs are often preferred for their ability to optimize motor efficiency across a wide range of operating conditions.

- These transmission devices play a crucial role in electric drive systems, enabling efficient power transmission, control of motor speed and torque, and adaptation to varying operating conditions.